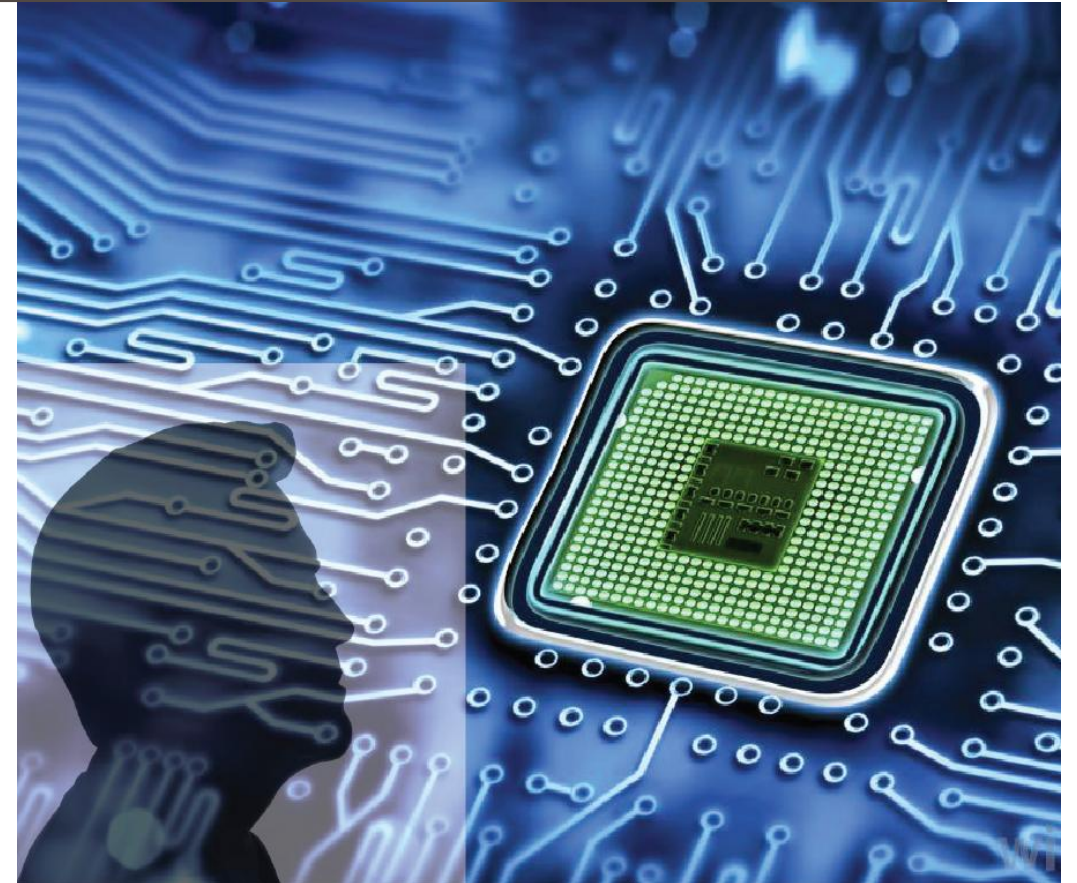


ADVANCED DIGITAL DESIGN AND VERIFICATION

Week-14 Lecture-2 Power Analysis and Optimization

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Power Analysis

Components of Power Dissipation

Power dissipation is broadly of two types:

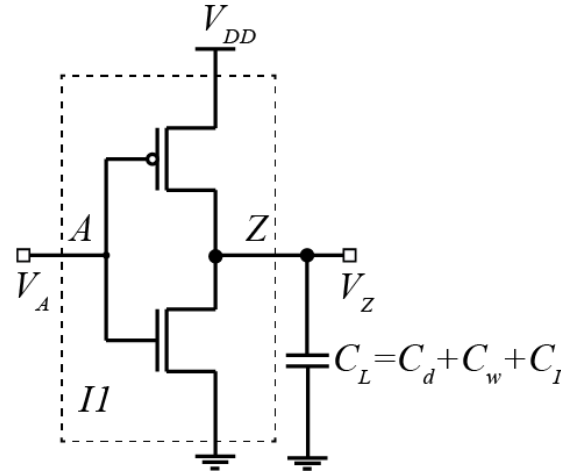
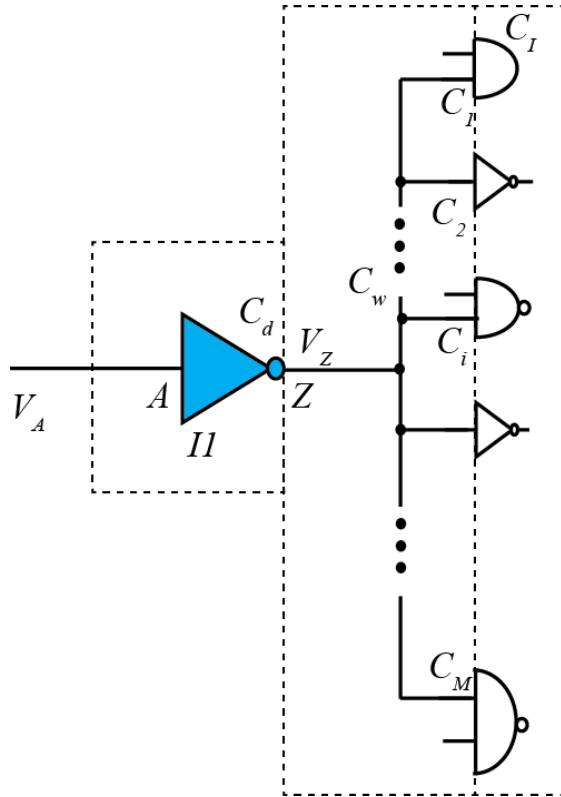
1. Dynamic Power Dissipation:

- Occurs when a circuit performs computation actively

2. Static Power Dissipation:

- When the circuit is powered on (supply voltages are applied), but it does not perform active computation

Dynamic Power Dissipation: Switching Power



- Energy dissipated in one cycle of $0 \rightarrow 1 \rightarrow 0$ transition:

$$E_{sw} = C_L V_{DD}^2$$

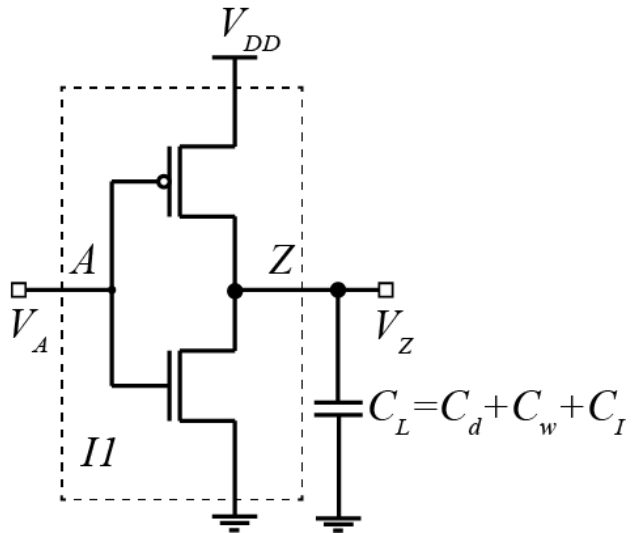
Switching power dissipated in a synchronous circuit:

$$P_{sw} = C_L V_{DD}^2 \alpha f_{clk}$$

where,

- f_{clk} = frequency of the clock in the circuit
- α = activity of the signal
 - define $\alpha = 1$ when the output completes one cycle of transition ($1 \rightarrow 0 \rightarrow 1$) in one clock period

Dynamic Power Dissipation: Short circuit Power



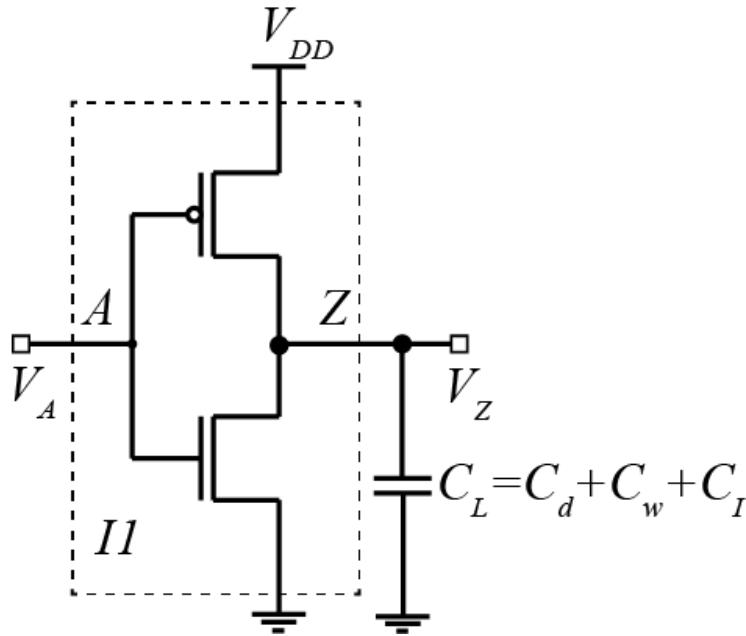
- Short circuit power dissipation:

$$P_{sc} = V_{DD} I_{SC}$$

- Power dissipated when short circuit condition occur:

$$P_{dyn} = P_{sw} + P_{sc}$$

Static Power Dissipation

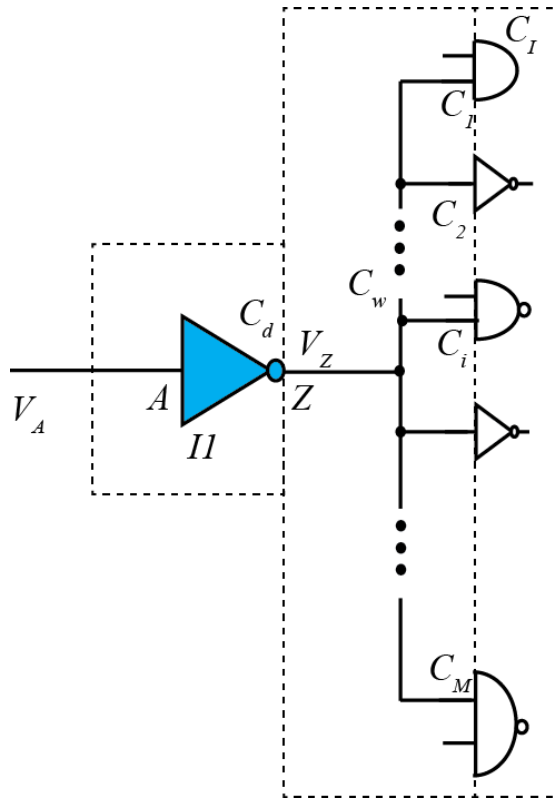


- Static power dissipation occurs because of:
 - Subthreshold current
 - Gate Leakage
 - Junction Leakage
- $P_{stat} = V_{DD}I_{leak}$

Total power dissipation in a circuit:

$$P_{tot} = P_{dyn} + P_{stat}$$

Technology Library Models: Dynamic Power



Energy dissipated in one cycle of $0 \rightarrow 1 \rightarrow 0$ transition:

$$E_{dyn} = C_L V_{DD}^2 + V_{DD} I_{SC} \tau_{SC}$$
$$E_{dyn} = (C_d + C_w + C_I) V_{DD}^2 + V_{DD} I_{SC} \tau_{SC}$$

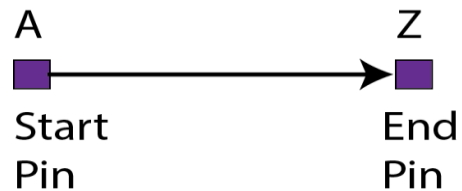
We can write:

$$E_{dyn} = C_d V_{DD}^2 + V_{DD} I_{SC} \tau_{SC} + (C_w + C_I) V_{DD}^2 = E_{int} + E_{ext}$$

- Energy dissipated inside a cell E_{int} is the property of the cell and modelled in the library
- Energy dissipated outside a cell E_{ext} depends on the environment (external load)
 - Tools can compute it after $(C_w + C_I)$ is known
- Power can be estimated using energy per transition by multiplying with activity and clock frequency

Non-linear Power Model (NLPM)

- Internal power dissipation depends on the output-load and input slew
- Modelled as two-dimensional table named **internal_power**
 - Referred to as Non-linear Power Model (NLPM)



- Rise and fall power can be represented as different arcs
- Values represent energy dissipated per transition

```
u_table_template(index_1) {  
    variable_1 : input_net_transition ;  
    variable_2 : total_output_net_capacitance ;  
    index_1( "10, 20, 30" ) ;  
    index_2( "1.2, 5.0, 15.0, 37.5" ) ;  
}  
...  
pin(Z) {  
    internal_power() {  
        related_pin : "A" ;  
        rise_power(index_1) {  
            values( "4, 5, 7, 12, ...3x4 table");  
        }  
        ...  
    }  
}
```


Technology Library Models: Static Power

- Static power dissipated inside a CMOS logic gate depends on the value (0 or 1) at its input pin
- Modeled using **when** condition in the library

```
cell (NAND2) { ...  
    cell_leakage_power : 125;  
    leakage_power () {  
        when : "!A & !B"; value : 20; }  
    leakage_power () {  
        when : "A & !B"; value : 150; }  
    leakage_power () {  
        when : "!A & B"; value : 200; }  
    leakage_power () {  
        when : "A & B"; value : 300; } ...
```

Estimating Power Dissipation

$$P_{tot} = C_L V_{DD}^2 \alpha f_{clk} + V_{DD} I_{SC} + V_{DD} I_{leak}$$

where,

- V_{DD} = supply voltage
- C_L = load capacitance
- f_{clk} = frequency of the clock in the circuit
- α = activity of the signal

- Computing power dissipation is a challenging problem.
 - Capacitance estimation
 - Accounting for the activity of signals

Activity of a signal depends on:

- Application being run on an IC
- Logical structure and the circuit topology

Power Optimization

Strategies for Reducing Power Dissipation

$$P_{tot} = C_L V_{DD}^2 \alpha f_{clk} + V_{DD} I_{SC} + V_{DD} I_{leak}$$

where,

- V_{DD} = supply voltage
- C_L = load capacitance
- f_{clk} = frequency of the clock in the circuit
- α = activity of the signal

Power saving strategies:

- Reduce supply voltage
- Reduce clock frequency
- Reduce activity of the signal
- Reduce load capacitance

Dynamic Voltage Frequency Scaling (DVFS)

- Utilizes the strategy of reducing supply voltage and clock frequency
- Popular technique employed in processors:
 - Exploit variations in workload to save energy
 - Full speed of a processor is utilized by only a few tasks or for a small time duration.
 - For the remaining period, the deadlines can be met at low speed and consuming significantly less energy

Illustration:

- Consider a processor that can perform a task in 10 ms at 1.2 GHz and 1.2 V.
- Let us reduce the clock frequency and the supply voltage to half (600 MHz, 0.6 V),
 - Task will now complete in double the original time, i.e., 20 ms.
 - Will reduce the switching power dissipation by 1/8 ($P_{tot} \propto V_{DD}^2 f_{clk}$)
 - Will reduce the energy consumption by 1/4

Power Gating

- Utilizes the strategy of eliminating supply voltage

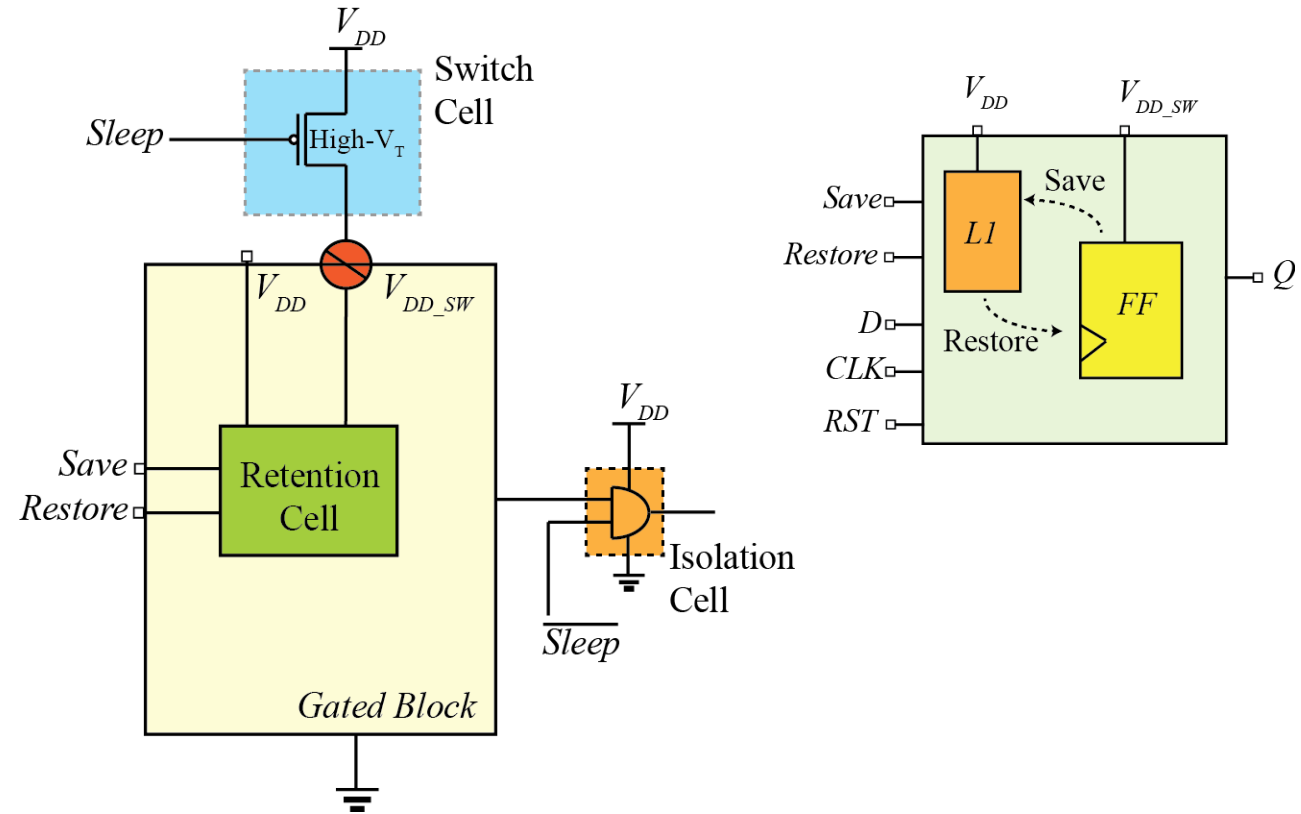
Power Gating:

- Switch off the power supply for a block
- Effective technique to tackle both static and dynamic components of power dissipation.
- Requires a careful circuit design and inserting specially designed circuit elements.

Power Gating: Circuit Elements

Circuit Elements

- Switch Cell
- Retention Cell
- Isolation Cell



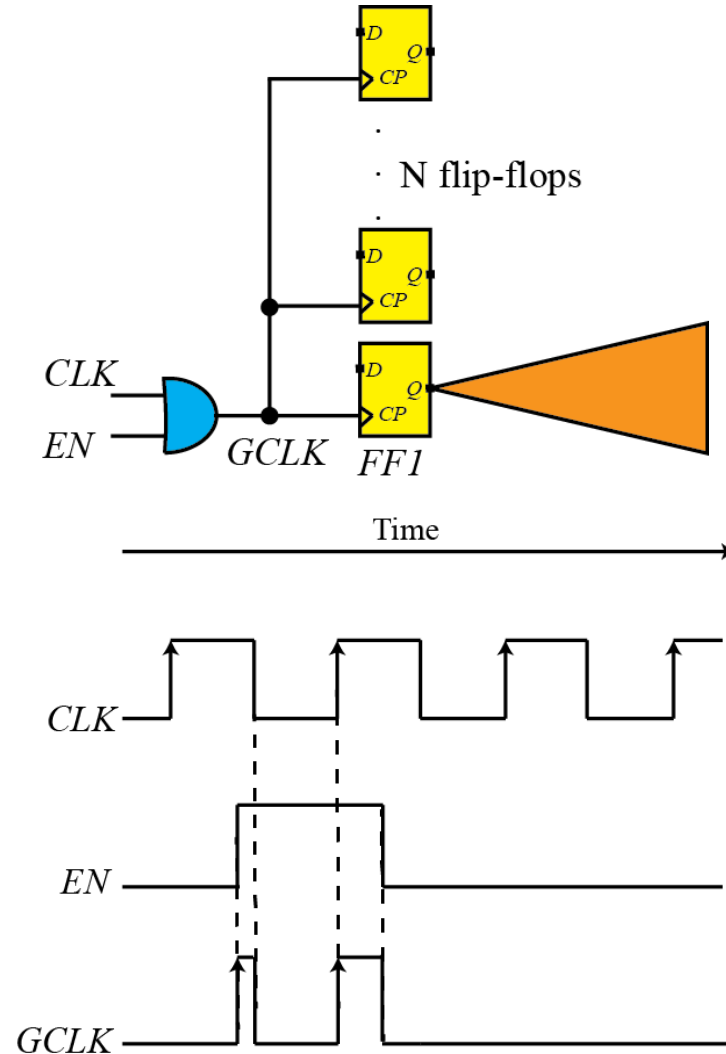
Clock Gating

- Utilizes the strategy of reducing activity

Assume that there is a set of N flip-flops that captures new data conditionally

- Shut off the clock when that condition is false
- Save power in charging/discharging capacitors in the clock network, including flip-flops
- Find enabling condition of clocking

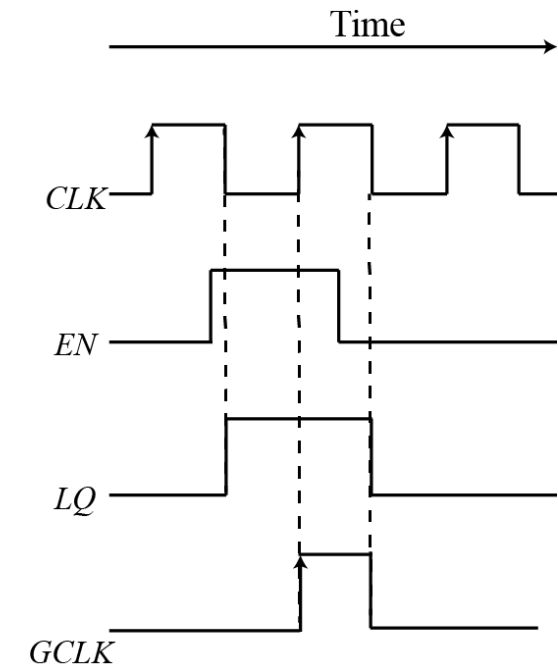
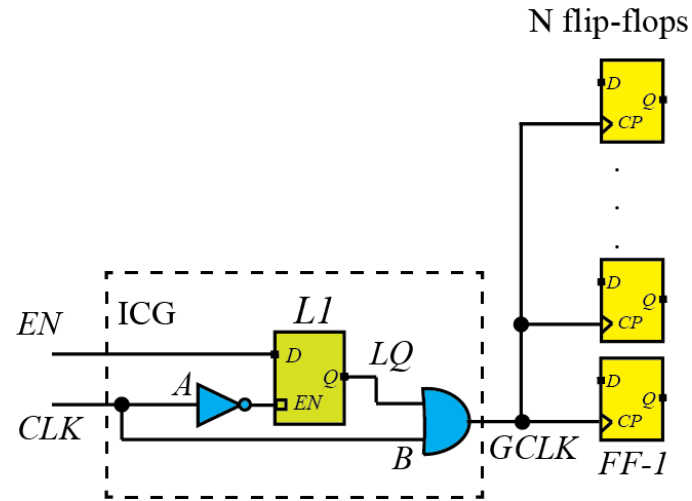
- Simple AND of EN and CLK will lead to glitch



Clock Gating: Integrated Clock Gater

Integrated Clock Gater (ICG)

- Logic gate with an enable signal so that clock is propagated when the $EN = 1$
- Latch (negative sensitive) allows EN to propagate only when CLK is low
- When CLK is high, Latch (LT) output is stable and glitch cannot propagate



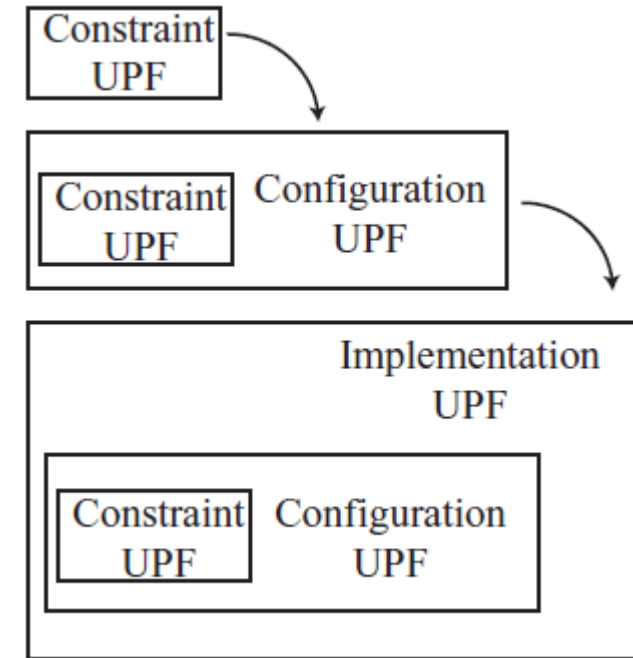
Power Intent: Unified Power Format (UPF)

Power Intent

- Power-related design specifications is known as **power intent**
- Power domains in the design, supply voltage levels, mapping of nets with voltage levels, power modes, and specially designed circuit elements for power management
- Specified using UPF

Unified Power Format (UPF)

- Commands to capture the power intent in a syntax consistent with the TCL
- UPF allows us to segregate the power intent from the logical intent captured in HDL
- UPF files can be successively refined in the design flow



References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.

Wishing you a bright future!